

A Primordial-Clock Candidate for de Sitter–MOND Gravity: Explicit Covariant Action, Derived Field Equations, a Dust-like Expanding Background, and an Obstruction to Instantaneous MOND on Expanding Initial Data

Carl P. Zimmerman
Briar Creek Tech

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Abstract

We record the exact current state of a candidate relativistic completion of Milgrom’s modified Newtonian dynamics (MOND) built on a primordial-clock foliation. What is established: an explicit covariant action with a clock scalar τ , a dynamical scalar χ , a cubic coupling $\gamma X \Box \chi$ and matter minimally coupled to the single metric; its varied field equations (metric, χ -current conservation, clock constraint, matter conservation); a reconstructed expanding FLRW background on which the clock sector is exactly pressureless dust; positive, luminal tensor modes ($c_T^2 = 1$); and a finite-wavelength scalar velocity matrix of rank one, with selected positivity results carrying Lean 4 certificates. What is not established, stated with equal weight: the coefficient functions were *reconstructed* to produce a chosen background rather than selected from first principles; the complete nonlinear constraint count is open; the recombination history is incomplete; exact galactic MOND, the measured G_N , the full post-Newtonian sector, nonlinear lensing, clusters and the cosmic microwave background are open; and the relation $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda}$ has not been derived from this action. On the solved spherical initial-data family the background-subtracted areal acceleration is bounded by $C(r)\varepsilon$, and the elementary inequality $1 - e^{-x} \leq x$ then shows that the exact exponential MOND law cannot hold instantaneously on that family (a Lean 4 certificate proves the inequalities assuming the force bound). The latest evolution-refinement test fails numerical convergence (momentum-constraint residual $7.41 \times 10^{-6} \rightarrow 1.77 \times 10^{-5}$ from 129 to 257 grid points). This is a status report, not a claim that the theory is established; the decisive next result is named.

1 Scope

This note fixes, in one place, what has actually been derived for the candidate and what has not. It is written so that every statement is either a theorem of the stated action, a computed quantity with its script named in the public repository, or an explicitly labelled open item. The construction was carried out collaboratively by AI systems under the author’s direction; nothing here is peer reviewed. An earlier flagship note in this series [9] argued for the cuscuton branch as the consistent structural class; the present candidate is the explicit attempt to realise a dynamical scalar sector

on a clock foliation, and it is reported with its failures. One correction to earlier statements in the programme is made explicit in Section 2.1.

2 The action

In units $c = 1$, signature $(-+++)$,

$$S = \int d^4x \sqrt{-g} \left[\frac{M^2}{2} (\mathcal{R} - 2\Lambda) + P(X, \tau) + sW(Y, \tau) - V(\tau) + \gamma X \square \chi \right] + S_m[g, \psi], \quad (1)$$

with

$$M^2 = \frac{1}{8\pi G_{\text{bare}}}, \quad X = -\nabla_\mu \chi \nabla^\mu \chi, \quad s = \sqrt{-\nabla_\mu \tau \nabla^\mu \tau} > 0, \quad n_\mu = -\frac{\nabla_\mu \tau}{s}, \quad (2)$$

$$h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu, \quad Q = n^\mu \nabla_\mu \chi, \quad D_\mu \chi = h_\mu{}^\nu \nabla_\nu \chi, \quad Y = D_\mu \chi D^\mu \chi = Q^2 - X. \quad (3)$$

Here τ defines the primordial-clock foliation, χ is the dynamical scalar sector, and γ is the cubic coupling (unrelated to γ_{PPN}). Ordinary matter couples minimally to the same metric.

2.1 Coefficient functions and the correction

The functions currently under test are

$$P(X, \tau) = -\frac{U}{2} \log\left(\frac{U - 2dX}{U - 2dq^2}\right) + 3\gamma q H_{\text{bg}}(X - q^2), \quad (4)$$

$$W(Y, \tau) = U + 2d\ell \left(\sqrt{1 + \frac{Y}{\ell}} - 1 \right) - 2\gamma q^2 \dot{q}_{\text{bg}}, \quad V(\tau) = U, \quad (5)$$

with prescribed background functions

$$\mathcal{A} = \frac{I}{a_{\text{bg}}^3}, \quad q = \frac{Q_c}{1+m}, \quad U = mq\mathcal{A}, \quad d = \frac{\mathcal{A}U}{2q(q\mathcal{A} + U)}, \quad \ell = \frac{q^2 m}{2}, \quad 3M^2 H_{\text{bg}}^2 = M^2 \Lambda + q\mathcal{A} + U. \quad (6)$$

During variation $q(\tau)$ and $H_{\text{bg}}(\tau)$ are prescribed functions of the clock; replacing them by local fields would change the action. The regular domain requires positive logarithm arguments and $1 + Y/\ell > 0$.

Correction. Earlier answers in the programme understated one point: these coefficient functions were reconstructed to produce a chosen background history. They have not been selected from first principles, and no statement below should be read as if they had. The coefficient history and differentiation rules are in `qwen_claude_field_theory/closure_2026/clock_response_repair_2026/nonlinear_infall_2026/backgro`

3 Field equations

Varying the metric,

$$M^2 (G_{\mu\nu} + \Lambda g_{\mu\nu}) = T_{\mu\nu}^m + T_{\mu\nu}^{\text{clock}}, \quad (7)$$

where, writing $\chi_\mu = \nabla_\mu \chi$,

$$T_{\mu\nu}^{\text{clock}} = 2P_X \chi_\mu \chi_\nu + sW n_\mu n_\nu - 2sW_Y D_\mu \chi D_\nu \chi + g_{\mu\nu}(P + sW - V) + 2\gamma(\Box\chi)\chi_\mu \chi_\nu + \gamma(\chi_\mu \nabla_\nu X + \chi_\nu \nabla_\mu X) - \gamma g_{\mu\nu} \nabla_\alpha \chi \nabla^\alpha X. \quad (8)$$

Varying χ gives an exactly conserved current,

$$\nabla_\mu J^\mu = 0, \quad J^\mu = -2(P_X + \gamma\Box\chi)\nabla^\mu \chi - \gamma\nabla^\mu X + 2sW_Y D^\mu \chi. \quad (9)$$

Varying the clock,

$$P_\tau + sW_\tau - V_\tau - \nabla_\mu(Wn^\mu - 2QW_Y D^\mu \chi) = 0, \quad (10)$$

which in clock gauge $\tau = t$ is the constraint

$$T \equiv V_\tau - P_\tau + WK - 2W_Y K^{ij} D_i \chi D_j \chi - 2QD_i(W_Y D^i \chi) = 0, \quad K_{ij} = \frac{1}{2}\mathcal{L}_n h_{ij}. \quad (11)$$

Its preservation $\partial_t T = 0$ yields an elliptic lapse equation on the regular branches studied. Matter's separate diffeomorphism invariance gives $\nabla_\mu T_m^{\mu\nu} = 0$ on the matter equations. The unrestricted spherical variation and its checks are documented in `.../clock_response_repair_2026/spherical_baryon_bridge/action/REPORT.md`.

4 Homogeneous background

For FLRW with proper-time derivatives,

$$\rho_{\text{clock}} = 2Q^2 P_X - P + V - 6\gamma H Q^3, \quad p_{\text{clock}} = P - V + sW(0, \tau) + 2\gamma Q^2 \dot{Q}, \quad j_{\text{clock}} = 2Q P_X - 6\gamma H Q^2, \quad (12)$$

and the background equations are

$$3M^2 H^2 = M^2 \Lambda + \rho_m + \rho_{\text{clock}}, \quad -2M^2 \dot{H} = \rho_m + p_m + \rho_{\text{clock}} + p_{\text{clock}}, \quad \frac{d}{dt}(a^3 j_{\text{clock}}) = 0, \quad V_\tau - P_\tau + 3HW(0, \tau) = 0. \quad (13)$$

The reconstructed solution has $j_{\text{clock}} = I/a^3$, $\rho_{\text{clock}} = Q_c I/a^3$, $p_{\text{clock}} = 0$: the clock supplies a dust-like background density. This is established for the constructed history only. It does not establish the required galaxy or cluster distribution of that component, nor the CMB spectrum. A same-action radiation calculation admits nearby expanding homogeneous solutions but has not produced a complete recombination history.

5 Perturbations

On the homogeneous branch the two tensor polarisations have

$$S_T^{(2)} = \frac{M^2}{8} \int dt d^3x a^3 \left[\dot{h}_{ij}^{TT} \dot{h}_{ij}^{TT} - \frac{\partial_k h_{ij}^{TT} \partial_k h_{ij}^{TT}}{a^2} \right], \quad (14)$$

so their kinetic energy is positive for $M^2 > 0$ and $c_T^2 = 1$. For finite nonzero wavenumber, after eliminating lapse and shift, the scalar velocity matrix is

$$K_{\text{red}} = \frac{a^3 \hat{B}}{2\Theta^2} \begin{pmatrix} q^2 & -qH \\ -qH & H^2 \end{pmatrix}, \quad \hat{B} = 2P_X + 4q^2 P_{XX} - 12\gamma H q + \frac{6\gamma^2 q^4}{M^2}, \quad \Theta = H + \frac{\gamma q^3}{M^2}, \quad (15)$$

of rank one on its regular branch: one dynamical scalar remains. Selected-background positivity results carry Lean 4 certificates with a symbolic bridge to the action. This does not prove the nonlinear count of two gravitational modes plus one healthy scalar, and the $k = 0$ sector needs its separate constraints (`.../clock_response_repair_2026/cubic_finite_wavelength/README.md`).

6 Spherical initial data and the obstruction to instantaneous MOND

For the solved spherical initial family $\rho_b(r) = \varepsilon\sigma(r)$, $\chi' = 0$, $Q = q$, $K^i_j = H\delta^i_j$, the Hamiltonian constraint gives

$$f(r) = A^{-2}(r) = 1 - \frac{1}{M^2 r} \int_0^r \rho_b(s) s^2 ds, \quad (16)$$

and preserving the clock constraint gives the lapse equation

$$S\Delta_h N - \lambda_0(N - 1) - \lambda_b \rho_b N = 0, \quad S = W - 2q^2 W_Y, \quad (17)$$

with coefficients derived from the action (`.../nonlinear_infall_2026/slice_action/REPORT.md`). The background-subtracted inward areal acceleration is

$$g_{\text{init}} = \frac{1-f}{2r} + \frac{r}{2M^2 N} \left[U(1-N) + 2\gamma q^2 (Q_t - \dot{q}_{\text{bg}}) \right], \quad (18)$$

and on the positive-coefficient branch the analysis establishes $0 \leq g_{\text{init}} \leq C(r)\varepsilon$. The exact exponential MOND law would require $(1 - e^{-g/a_0})g = g_N$ with $g_N = \varepsilon n(r)$; the inequality $1 - e^{-x} \leq x$ gives

$$0 \leq \frac{(1 - e^{-g_{\text{init}}/a_0}) g_{\text{init}}}{\varepsilon n(r)} \leq \frac{C(r)^2 \varepsilon}{a_0 n(r)} \longrightarrow 0, \quad (19)$$

whereas MOND requires the ratio to equal one. This excludes instantaneous MOND on this expanding initial-data family. A late-time galactic branch remains uncomputed. The Lean certificate (`.../nonlinear_infall_2026/lean/SmallSourceObstruction.lean`) proves the inequalities assuming the force bound; the PDE argument supplying that bound is outside Lean.

7 Numerical status

The action, initial-data, potential, full-gradient, background, scalar-principal, tensor and seven Lean checks exit 0. The latest evolution-refinement test exits 1: increasing the grid from 129 to 257 points increased the momentum-constraint residual from 7.41×10^{-6} to 1.77×10^{-5} . That is an unresolved numerical validation failure. It is not a proof that the theory is impossible, and it is not evidence that it works.

8 Status

Requirement	Current evidence
Explicit action and varied equations	Established
Separate ordinary-matter conservation	Established on the matter equations
Expanding FLRW	Constructed background; limited radiation extension
Positive, luminal tensor modes	Derived on homogeneous clock backgrounds
Scalar health	Partial linear results; conditional Lean certificates
Complete nonlinear constraint count	Open
$\Phi = \Psi$	Independently supported at linear order on the tested family
Exact galactic MOND and measured G_N	Open; instantaneous MOND excluded on the expanding initial family
Full PPN, nonlinear lensing, CMB, clusters	Open
First-principles coefficients and $\kappa = \frac{1}{2}$	Open; coefficients reconstructed, $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda}$ not derived
Converged nonlinear evolution	Failed at the latest refinement

9 Relation to the independent gate analysis

A separate programme in the same repository (`fable_independent_2026`, lanes L129–L176, a Lean 4 file of 99 theorems with zero `sorry`) tests candidate architectures against the galaxy, Solar-System, forest, growth and CMB gates independently of any construction. Two of its results bear directly on this candidate and are stated here so that the two lines are not read as contradicting each other. First, a necessity certificate: any theory on these equations passing all gates must carry a dark fraction that rises with host mass (approximately $f \propto M^{0.16}$ from spirals to clusters, with $f \geq 0.988$ required at recombination and $f \lesssim 0.1$ tolerated in spirals), a non-barotropic effective fluid, and a locally screened preferred-frame source. Second, every mechanism so far computed for that mass dependence (thermal relic, two-body decay, velocity kick, clock-switched sound speed, potential-depth threshold) fails on the Lyman- α forest, on small-scale cosmic shear, or on N_{eff} ; and it was found that all of those verdicts assume the dark component alone must reproduce Λ CDM’s small-scale power, whereas a kernel-boosted baryonic sector also builds structure there. The framework’s own nonlinear $P(k, z)$ with the kernel active is being computed as the baseline that those verdicts require. The present candidate’s dust-like clock background is exactly the component the necessity certificate constrains; whether its distribution can be mass dependent is part of what is open.

10 What would decide

The next discriminating result is a converged baryon-sourced evolution of these equations, or a directly constructed quasistatic branch, showing whether they produce the required MOND force and the observed lensing at late times. Until one of those exists, this candidate is an explicit, varied,

partially certified action with a working expanding background and named failures, and nothing more.

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